



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - JAN/FEB 2023

SUBJECT : ECE 2154 - NETWORK ANALYSIS
(MAKEUP)

NETWORK ANALYSIS [ECE 2154]

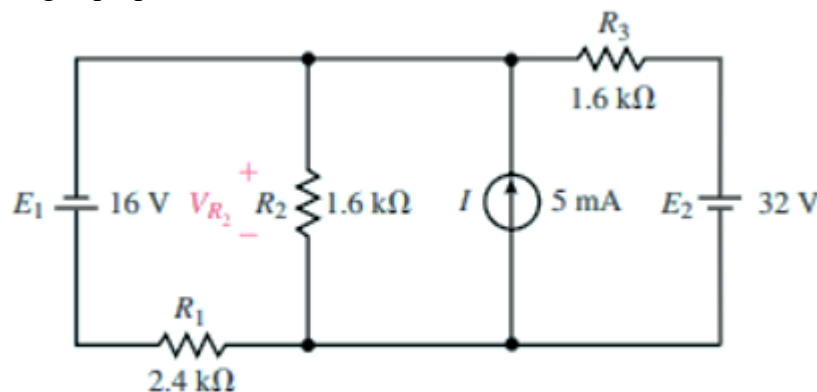
Marks: 50

Duration: 180 mins.

A

Answer all the questions.

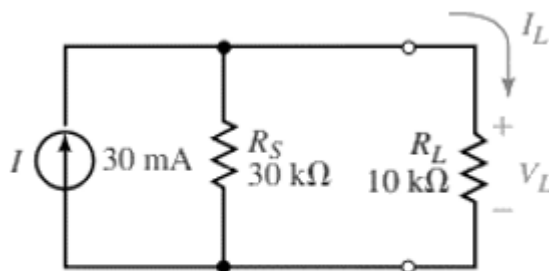
- 1A) Determine the voltage drop across the resistor R_2 of the circuit shown in Figure 1A. using superposition theorem.



(4)

Figure 1A

- 1B) Convert the current source of Figure 1B into a voltage source and verify that the voltage, V_L , across the load is the same for each source.



(3)

Figure 1B

- 1C) (3)

Find the current supplied by the battery in the network of Figure 1C.

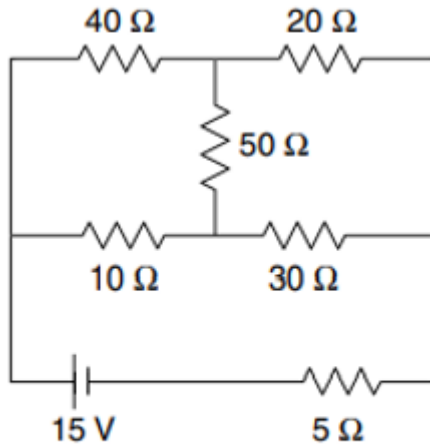


Figure 1C.

- 2A) In the network of Fig. 2A, the switch is changed from position 'a' to 'b' at $t = 0$. Solve for i , di/dt and d^2i/dt^2 at $t = 0^+$.

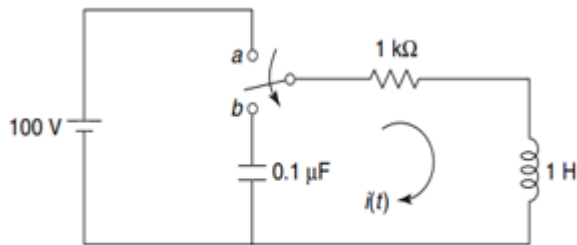


Fig. 2A

- 2B) In the network shown in the Fig. 2B, the switch is moved from position a to b at $t = 0$, having been in position a for a long time before $t = 0$. The capacitor C_2 is uncharged at $t = 0$. Find $i(t)$ and $V_2(t)$ for $t > 0$.

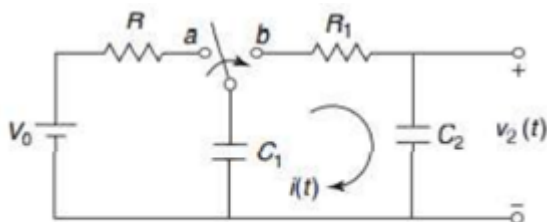


Fig. 2B

- 2C) For the differential equation given below, find the particular solution for $v(t)$. $d^2v/dt^2 + 5dv/dt + 6v = e^{-2t} + 5e^{-3t}$. (3)
- 3A) A 10V step input is applied to a high-pass RC circuit. Draw to scale the output waveform for the cases: (a) $RC=1\text{ms}$ and $t=0,1,2,4\text{ms}$ (b) $RC=10\text{ms}$ and $t=0,1,2,4,10,20,40\text{ms}$ (c) $RC=100\text{ms}$ and $t=0,1,2,4,10,20,40,100\text{ms}$. (4)
- 3B) A limited ramp shown in Figure 3B is applied to an RC high-pass circuit. The time constant of the RC circuit is 2ms. Calculate the maximum value of the output voltage (3)

and the output at the end of the input waveform.

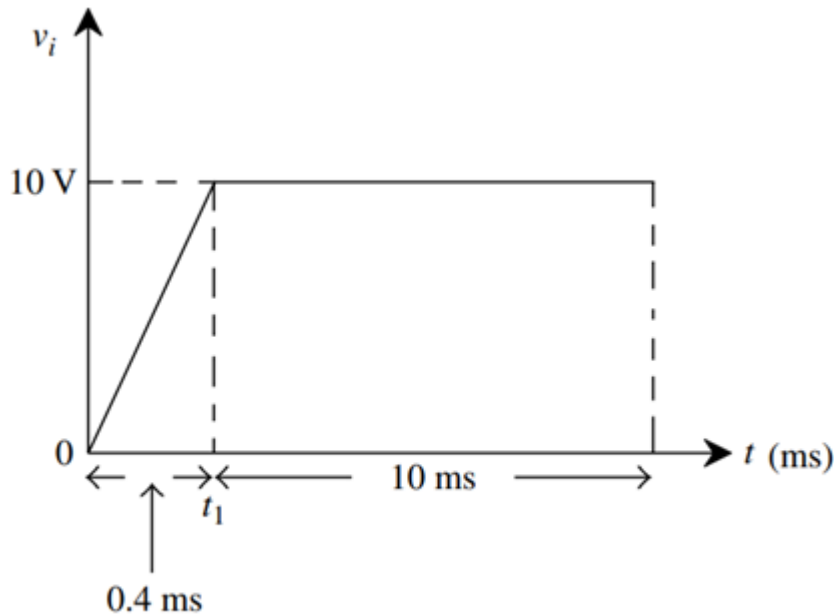


Figure 3B

- 3C) Sketch the output waveform for an RC integrating circuit for a pulse input of amplitude V when:
 (a) $\tau = 10t_p$, (b) $\tau = t_p$, (c) $\tau = 0.1t_p$. (3)

- 4A) For the network shown in the Fig. 4A, the switch is closed at $t = 0$. Determine the current $i(t)$ assuming zero initial conditions using Laplace Transform.

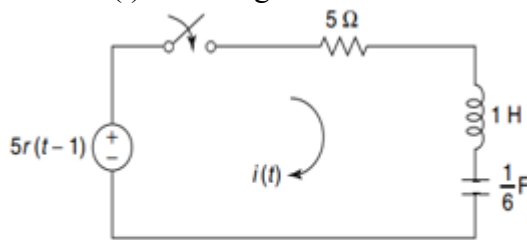


Fig. 4A

- 4B) For the network shown in the Fig. 4B, determine the current $i(t)$ when the switch is closed at $t=0$ with zero initial conditions using Laplace Transform.

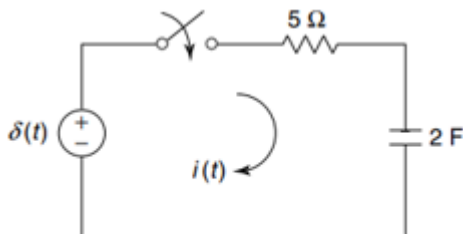


Fig 4B

- 4C) For the network shown in the Fig. 4C, the switch is closed at $t=0$. Determine the current $i(t)$ assuming zero initial conditions in the network using Laplace Transform. (3)

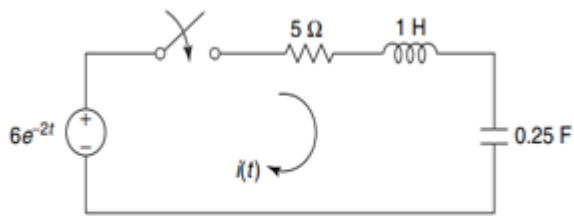


Fig. 4C

5A) Find the transmission parameters for the two-port network shown in Figure 5A.

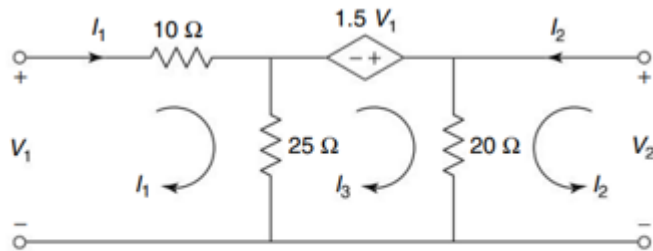


Figure 5A

5B) For the resistive bridged T network shown in Figure 5B, find $\frac{V_2}{V_1}, \frac{V_2}{I_1}, \frac{I_2}{V_1}, \frac{I_2}{I_1}$.

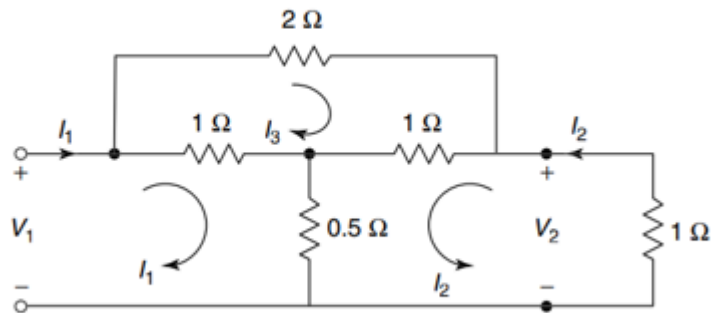


Figure 5B

5C) Derive the expression for z parameters in terms of ABCD parameters.

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